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Touch display apparatus

Abstract

A touch display apparatus includes a display panel, first light-emitting elements and a first conductive layer. The first light-emitting elements are disposed on a first auxiliary display area of the display panel. The first conductive layer is disposed on the display panel and includes first mesh-shaped touch electrodes and first dummy electrodes. Each of the first dummy electrodes is disposed between two adjacent first mesh-shaped touch electrodes and is structurally separated from the two adjacent first mesh-shaped touch control electrodes. At least one portion of the first dummy electrode is electrically connected to at least one of the first light-emitting elements disposed in the first auxiliary display area.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2023/0341984	12/2022	Pang et al.	N/A	N/A
2024/0176454	12/2023	Hasegawa	N/A	G06F 3/041

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
115206197	12/2021	CN	N/A
115485655	12/2021	CN	N/A
115756209	12/2022	CN	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims the priority benefit of Taiwan application serial no. 113104439, filed on Feb. 5, 2024. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

(2) The disclosure relates to a display apparatus, and in particular to a touch display apparatus.

Description of Related Art

(3) In order to meet the needs of users for displaying various shared information and comprehensive information in a large area, spliced display devices with multiple display panels integrated have been widely used in various fields. For example, spliced display devices are commonly used in public information display areas (such as large advertisements, monitoring and management screens, etc.). However, spliced display devices usually have the problem of discontinuity in the overall display image, which affects the viewing quality. Therefore, light-emitting elements can be provided on at least one side of the auxiliary display area of the display panel to eliminate joints in the spliced display device. However, the arrangement of circuits for driving the light-emitting elements will make the overall thickness of the display device thicker and complicate the manufacturing process.

SUMMARY

- (4) This disclosure provides a touch display apparatus with good performance.
- (5) A touch display apparatus of this disclosure includes a display panel, a plurality of first light-emitting elements and a first conductive layer. The display panel has a main display area and a first auxiliary display area outside the main display area. The first light-emitting elements are arranged on the first auxiliary display area of the display panel. The first conductive layer is disposed on the display panel and comprises a plurality of first mesh-shaped touch electrodes and a plurality of first dummy electrodes. The first mesh-shaped touch electrodes are located on the main display area of the display panel. The first dummy electrodes are located on the main display area of the display panel. Each of the first dummy electrodes is disposed between two adjacent first mesh-shaped touch electrodes and structurally separated from the two adjacent first mesh-shaped touch electrodes. At least one portion of each of the first dummy electrodes is electrically connected to at least one of the first light-emitting elements disposed in the first auxiliary display area.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a top view of a touch display apparatus according to an embodiment of the present disclosure.
- (2) FIG. 2 is a schematic cross-sectional view of a touch display apparatus according to an embodiment of the present disclosure.
- (3) FIG. 3 is a top view and enlarged schematic diagram of the first mesh-shaped touch electrode, first dummy electrode, second mesh-shaped touch electrode and second dummy electrode according to one embodiment of this disclosure.
- (4) FIG. 4 is a partially enlarged schematic diagram of the first conductive layer of an embodiment of the present disclosure.
- (5) FIG. 5 is a partial enlarged schematic diagram of the second conductive layer of an embodiment of the present disclosure.
- (6) FIG. 6 is a schematic top view of the first mesh-shaped touch electrode, first dummy electrode, first light-emitting element and first signal line of the touch display apparatus according to another embodiment of the present disclosure.
- (7) FIG. 7 is a schematic top view of the first mesh-shaped touch electrode, first dummy electrode, first light-emitting element and first signal line of the touch display apparatus according to another embodiment of the present disclosure.
- (8) FIG. 8 is a schematic top view of the second mesh-shaped touch electrode, second dummy electrode, second light-emitting element and second signal line of the touch display apparatus according to yet another embodiment of the present disclosure.

DESCRIPTION OF THE EMBODIMENTS

- (9) Reference will now be made in detail to exemplary embodiments provided in the disclosure, examples of which are illustrated in accompanying drawings. Wherever possible, identical reference numerals are used in the drawings and descriptions to refer to identical or similar parts.
- (10) It should be understood that when a device such as a layer, film, region or substrate is referred to as being “on” or “connected to” another device, it may be directly on or connected to another device, or intervening devices may also be present. In contrast, when a device is referred to as being “directly on” or “directly connected to” another device, there are no intervening devices present. As used herein, the term “connected” may refer to physical connection and/or electrical connection. Besides, if two devices are “electrically connected” or “coupled”, it is possible that other devices are present between these two devices.
- (11) The term “about,” “approximately,” or “substantially” as used herein is inclusive of the stated

value and a mean within an acceptable range of deviation for the particular value as determined by people having ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity (i.e., the limitations of the measurement system). For example, “about” may mean within one or more standard deviations, for example, $\pm 30\%$, $\pm 20\%$, $\pm 10\%$, or $\pm 5\%$ of the stated value. Moreover, a relatively acceptable range of deviation or standard deviation may be chosen for the term “about,” “approximately,” or “substantially” as used herein based on optical properties, etching properties or other properties, instead of applying one standard deviation across all the properties.

(12) Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by people of ordinary skill in the art. It will be further understood that terms, such as those defined in the commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and the invention and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

(13) FIG. 1 is a top view of a touch display apparatus according to an embodiment of the present disclosure. FIG. 2 is a schematic cross-sectional view of a touch display apparatus according to an embodiment of the present disclosure. FIG. 3 is a top view and enlarged schematic diagram of the first mesh-shaped touch electrode, first dummy electrode, second mesh-shaped touch electrode and second dummy electrode according to one embodiment of this disclosure.

(14) It should be noted that FIG. 1 uses multiple arrows to represent the first mesh-shaped touch electrode **410**, the first dummy electrode **420**, the second mesh-shaped touch electrode **610** and the second dummy electrode **620** to concisely express the relative relationship between the first mesh-shaped touch electrodes **410**, the first dummy electrodes **420**, the second mesh-shaped touch electrodes **610**, the second dummy electrodes **620**, the first light-emitting elements **200** and the second light-emitting elements **300**. However, the shapes of the first mesh-shaped touch electrode **410**, the first dummy electrode **420**, the second mesh-shaped touch electrodes **610** and the second dummy electrodes **620** are not the multiple arrows shown in FIG. 1. The shapes of the first mesh-shaped touch electrode **410**, the first dummy electrode **420**, the second mesh-shaped touch electrodes **610** and the second dummy electrodes **620** may be referred to FIG. 3.

(15) Referring to FIG. 1, FIG. 2 and FIG. 3, the touch display apparatus **10** includes a display panel **100**. The display panel **100** has a main display area **100a** and an auxiliary display area **100b** outside the main display area **100a**. In one embodiment, the display panel **100** has pixel electrodes arranged in a unit cell (cell), the main display area **100a** may refer to an area occupied by the pixel electrodes, and the auxiliary display area **100b** may refer to an area of the display panel **100** not occupied by the pixel electrodes. In one embodiment, the auxiliary display area **100b** includes a first auxiliary display area **100b-1**, wherein the main display area **100a** and the first auxiliary display area **100b-1** are arranged in the first direction **d1**. In one embodiment, the auxiliary display area **100b** may optionally further include a second auxiliary display area **100b-2**, and the main display area **100a** and the second auxiliary display area **100b-2** are arranged in the second direction **d2**, wherein the first direction **d1** and the second direction **d2** are intersected. For example, in one embodiment, the first auxiliary display area **100b-1** may be a straight strip area located on the right or left side of the main display area **100a**, the second auxiliary display area **100b-2** may be a horizontal strip area located on the upper or lower side of the main display area **100a**, the first auxiliary display area **100b-1** and the second auxiliary display area **100b-2** may form an L-shaped area, but this disclosure is not limited to thereto. In one embodiment, the display panel **100** is, for example, a liquid crystal display panel, but this disclosure is not limited to thereto.

(16) Referring to FIG. 1, the touch display apparatus **10** further includes first light-emitting elements **200** arranged on the first auxiliary display area **100b-1** of the display panel **100**. In one embodiment, the first light-emitting element **200** is, for example, a micro light-emitting diode (μ LED), but this disclosure is not limited to thereto. In one embodiment, the touch display

apparatus **10** may also optionally include second light-emitting elements **300** disposed on the second auxiliary display area **100b-2** of the display panel **100**. In one embodiment, the second light-emitting element **300** is, for example, a micro light-emitting diode (μ LED), but this disclosure is not limited to thereto. When the touch display apparatus **10** is spliced with other display apparatus/other touch display apparatus to form a spliced display apparatus (not shown) with a large display screen, a light beam emitted by the first light-emitting element **200** and/or the second light-emitting element **300** can be used to eliminate the splicing seams between the touch display apparatus **10** and other display apparatus/other touch display apparatus to improve the picture quality of the spliced display apparatus.

(17) Referring to FIGS. **1** and **2**, the touch display apparatus **10** further includes a first conductive layer **400** disposed on the display panel **100**. In one embodiment, the touch display apparatus **10** further includes at least one insulation layer **500** and a second conductive layer **600**, wherein at least one insulation layer **500** is disposed on the first conductive layer **400**, and the second conductive layer **600** is disposed on at least one insulation layer **500**. In one embodiment, the first conductive layer **400** and the second conductive layer **600** are made of metal, for example, but this disclosure is not limited to thereto.

(18) In one embodiment, the touch display apparatus **10** may further selectively include a dielectric layer **700** and a transparent protective cover **800**. The dielectric layer **700** is disposed on the second conductive layer **600**, and the transparent protective cover **800** is disposed on the dielectric layer **700**. In one embodiment, the dielectric layer **700**, the second conductive layer **600**, at least one insulation layer **500** and the first conductive layer **400** may be sequentially formed on the transparent protective cover **800** to form a touch substrate TP, and the touch substrate TP may be attached to the outer surface **100s** of the display panel **100** through the optical adhesive **900**. However, this disclosure is not limited to thereto. In another embodiment not shown, the first conductive layer **400**, at least one insulation layer **500**, the second conductive layer **600** and a protective layer (not shown) may be sequentially formed on the outer surface **100s** of the display panel **100** so as to form a touch panel on cell structure.

(19) Referring to FIGS. **1**, **2** and **3**, the first conductive layer **400** includes first mesh-shaped touch electrodes **410** and first dummy electrodes **420** located on the main display area **100a** of the display panel **100**, wherein each of the first dummy electrode **420** is disposed between two adjacent first mesh-shaped touch electrodes **410** and is structurally separated from the two adjacent first mesh-shaped touch electrodes **410**. Each of the first mesh-shaped touch electrodes **410** extends substantially in a first direction **d1**, and the first mesh-shaped touch electrodes **410** are arranged in a second direction **d2** intersected with the first direction **d1**.

(20) It is worth noting that at least one portion of the first dummy electrode **420** is electrically connected to at least one first light-emitting element **200** disposed in the first auxiliary display area **100b-1**. In other words, in addition to improving the visual effect, the first dummy electrode **420** can also be used as a circuit to drive the first light-emitting element **200**. Thereby, while eliminating splicing seams by using the first light-emitting element **200** located in the first auxiliary display area **100b-1**, the touch display apparatus **10** does not need to add additional film layers for setting up circuits for driving the first light-emitting element **200**, the thickness **T** of the touch display apparatus **10** can be reduced, and the manufacturing process of the touch display apparatus **10** can be simplified.

(21) FIG. **4** is a partially enlarged schematic diagram of the first conductive layer of an embodiment of the present disclosure. FIG. **4** corresponds to local **R1** of FIG. **3**. Referring to FIGS. **1**, **3** and **4**, in one embodiment, the first dummy electrode **420** disposed between two adjacent first mesh-shaped touch electrodes **410** may include first conductive portions **422** structurally separated from each other, and the first conductive portions **422** of the first dummy electrode **420** are electrically connected to the first light-emitting elements **200** disposed in the first auxiliary display area **100b-1**. In one embodiment, the touch display apparatus **10** may further optionally include a

first driving element COF1, one end of each of the first conductive portions 422 may be electrically connected to the first driving element COF1, and the other end of each of the first conductive portions 422 may be electrically connected to a corresponding at least one first light-emitting element 200. In one embodiment, the first driving element COF1 may include a chip, and the chip may be bonded to the display panel 100 through a Chip On Film (COF) package, but this disclosure is not limited to thereto.

(22) Referring to FIG. 3 and FIG. 4, in one embodiment, each of the first conductive portion 422 may include first line segments 422a and second line segments 422b, the first line segments 422a and the second line segments 422b are alternately arranged and connected in series with each other, the first line segments 422a are substantially parallel to each other and extend in a third direction d3, the first direction d1 and the second direction d2 are substantially perpendicular, the third direction d3, the first direction d1 and the second direction d2 are intersected, and the second line segments 422b are substantially parallel to each other and extend in a fourth direction d4, and the fourth direction 4 intersects with the first direction d1, the second direction d2 and the third direction d3. In one embodiment, the first line segments 422a and the second line segments 422b of each of the first conductive portion 422 may be connected to form a zigzag-shaped wire.

(23) Referring to FIGS. 3 and 4, in one embodiment, the first conductive portions 422 of the first dummy electrode 420 define first dummy grids MS1'. Specifically, in one embodiment, a first line segment 422a of a first conductive portion 422 is disposed opposite a first line segment 422a of another adjacent first conductive portion 422, and a second line segment 422b of the first conductive portion 422 is disposed opposite a second line segment 422b of the another adjacent first conductive portion 422 to define a first dummy grid MS1'.

(24) In one embodiment, a shape of the first dummy grid MS1' defined by the two adjacent first conductive portions 422 of the first dummy electrode 420 is substantially the same as a shape of the first grid MS1 of the first mesh-shaped touch electrode 410, and a size of the first dummy grid MS1' is substantially the same as a size of the first grid MS1 of the first mesh-shaped touch electrode 410. For example, in one embodiment, the shapes of the first dummy grid MS1' and the first grid MS1 may both be rhombus, and lengths of the two diagonals L1' and L2' of the rhombus-shaped first dummy grid MS1' are substantially equal to lengths of the two diagonals L1 and L2 of the rhombus-shaped first grid MS1 respectively.

(25) Referring to FIG. 1 and FIG. 3, in one embodiment, the second conductive layer 600 includes second mesh-shaped touch electrodes 610 and second dummy electrodes 620 disposed on the main display area 100a of the display panel 100, the second mesh-shaped touch electrodes 610 and the first mesh-shaped touch electrodes 410 are intersected, each of the second dummy electrode 620 is disposed between two adjacent second mesh-shaped touch electrodes 610, each of the second dummy electrode 620 and the two adjacent second mesh-shaped touch electrodes 610 are structurally separated from each other.

(26) It is worth noting that at least one portion of the second dummy electrode 620 is electrically connected to at least one second light-emitting element 300 disposed in the second auxiliary display area 100b-2. In other words, in addition to improving the visual effect, the second dummy electrode 620 can also be used as a circuit to drive the second light-emitting element 300. Thereby, while using the second light-emitting element 300 located in the second auxiliary display area 100b-2 to eliminate splicing seams, the touch display apparatus 10 does not need to add additional film layers for setting up circuits for driving the second light-emitting element 300, the thickness T of the touch display apparatus 10 is not too thick, and the manufacturing process of the touch display apparatus 10 can be simplified.

(27) FIG. 5 is a partial enlarged schematic diagram of the second conductive layer of an embodiment of the present disclosure. FIG. 5 corresponds to local R2 in FIG. 3. Referring to FIGS. 1, 3 and 5, the second conductive layer 600 includes second mesh-shaped touch electrodes 610 and second dummy electrodes 620 located on the main display area 100a of the display panel 100,

wherein each of the second dummy electrodes **620** is disposed on two adjacent second mesh-shaped touch electrodes **610**, and the each of the second dummy electrodes **620** and the two adjacent second mesh-shaped touch electrodes **610** are structurally separated from each other. Each of the second mesh-shaped touch electrodes **610** substantially extends in the second direction **d2**, and the second mesh-shaped touch electrodes **610** are arranged in the first direction **d1**. One of the first mesh-shaped touch electrode **410** and the second mesh-shaped touch electrode **610** is a driving electrode (Tx), and the other of the first mesh-shaped touch electrode **410** and the second mesh-shaped touch electrode **610** is a receiving electrode (Rx).

(28) In one embodiment, the second dummy electrode **620** disposed between two adjacent second mesh-shaped touch electrodes **610** may include second conductive portions **622** that are structurally separated from each other, and the second conductive portions **622** of the second dummy electrode **620** are electrically connected to the second light-emitting elements **300** disposed in the second auxiliary display area **100b-2**. In one embodiment, the touch display apparatus **10** may further optionally include a second driving element COF2, one end of each of the second conductive portion **622** may be electrically connected to the second driving element COF2, and the other end of the each of the second conductive portion **622** may be electrically connected to the corresponding a least one second light-emitting element **300**. In one embodiment, the second driving element COF2 may include a chip, and the chip may be bonded to the display panel **100** through a Chip On Film (COF) package, but this disclosure is not limited to thereto.

(29) Referring to FIG. 3 and FIG. 5, in one embodiment, each of the second conductive portion **622** may include first line segments **622a** and second line segments **622b**, the first line segments **622a** and the second line segments **622b** are alternately arranged and connected in series with each other, the first line segments **622a** are substantially parallel to each other and extend in the third direction **d3**, and the second line segments **622b** are substantially parallel to each other and extend in the fourth direction **d4**. In one embodiment, the first line segments **622a** and the second line segments **622b** of each of the second conductive portion **622** may be connected to form a zigzag-shaped wire.

(30) Referring to FIGS. 3 and 5, in one embodiment, the second conductive portions **622** of the second dummy electrode **620** define second dummy grids MS2'. Specifically, in one embodiment, a first line segment **622a** of a second conductive portion **622** is disposed opposite a first line segment **622a** of another adjacent second conductive portion **622**, and a second line segment **622b** of the first conductive portion **422** is disposed opposite a second line segment **622b** of the another adjacent first conductive portion **422** to define a second dummy grid MS2'.

(31) In one embodiment, a shape of the second dummy grid MS2' defined by the two adjacent second conductive portions **622** of the second dummy electrode **620** is substantially the same as a shape of the second grid MS2 of the second mesh-shaped touch electrode **610**, and a size of the second dummy grid MS2' is the same as a size of the second grid MS2 of the second mesh-shaped touch electrode **610** is substantially the same. For example, in one embodiment, the shapes of second dummy grid MS2' and the second grid MS2 may both be rhombus, and lengths of the two diagonals L3' and L4' of the rhombus-shaped second dummy grid MS2' are substantially equal to lengths of the two diagonals L3 and L4 of the rhombus-shaped second grid MS2 respectively.

(32) Referring to FIG. 3, in one embodiment, n first conductive portions **422** are disposed between two adjacent first mesh-shaped touch electrodes **410**, and m second conductive portions **422** are disposed between two adjacent second mesh-shaped touch electrodes **610**. wherein n and m are positive integers greater than or equal to 1. For example, in one embodiment, n=3 and m=4. However, this disclosure is not limited to thereto. The values of n and m may be adjusted depending on the resolution.

(33) In the following embodiment, the reference numerals and part of the description of the foregoing embodiment are applied, where the same reference numerals are used to indicate the same or similar components, and descriptions of the same technical contents are omitted. Reference may be made to the foregoing embodiment for the omitted descriptions, which will not be repeated

in following embodiment.

(34) FIG. 6 is a schematic top view of the first mesh-shaped touch electrode, first dummy electrode, first light-emitting element and first signal line of the touch display apparatus according to another embodiment of the present disclosure. The touch display apparatus **10A** of FIG. 6 is similar to the aforementioned touch display apparatus **10**. The difference between the two is that in the embodiment of FIG. 6, a first conductive portion **422** of the first dummy electrode **420** may be electrically connected to a plurality of first light-emitting elements **200**, first ends **201** of the first light-emitting element **200** are electrically connected to first signal lines **VSS1** respectively, and second ends **202** of the first light-emitting elements **200** are electrically connected to the same first conductive portion **422**. That is to say, when the resolution of the touch display apparatus **10B** is improved, the first light-emitting elements **200** may share a first conductive portion **422** so as to simplify the circuit design for driving the first light-emitting element **200**, wherein the number of the first light-emitting elements **200** sharing the same first conductive portion **422** depends on actual needs.

(35) FIG. 7 is a schematic top view of the first mesh-shaped touch electrode, first dummy electrode, first light-emitting element and first signal line of the touch display apparatus according to another embodiment of the present disclosure. The touch display apparatus **10B** of FIG. 7 is similar to the touch display apparatus **10A** of FIG. 6. The difference between the two is that in the embodiment of FIG. 6, three first light-emitting elements **200** share one first conductive portion **422**; in the embodiment of FIG. 7, six first light-emitting elements **200** share one first conductive portion **422**.

(36) FIG. 8 is a schematic top view of the second mesh-shaped touch electrode, second dummy electrode, second light-emitting element and second signal line of the touch display apparatus according to yet another embodiment of the present disclosure. The touch display apparatus **10C** of FIG. 8 is similar to the aforementioned touch display apparatus **10**. The difference between the two is that in the embodiment of FIG. 8, a second conductive portion **622** of the second dummy electrode **620** may be electrically connected to second light-emitting elements **300**, first ends **301** of the second light-emitting element **300** are electrically connected to second signal lines **VSS2** respectively, and second ends **302** of the second light-emitting elements **300** are electrically connected to the same second conductive portion **622**. That is to say, when the resolution of the touch display apparatus **10C** is improved, the second light-emitting elements **300** may share a second conductive portion **622** so as to simplify the circuit design for driving the second light-emitting element **300**, wherein the number of the second light-emitting elements **300** sharing the same second conductive portion **622** depends on actual needs.

(37) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed embodiments without departing from the scope or spirit of the disclosure. In view of the foregoing, it is intended that the disclosure covers modifications and variations provided that they fall within the scope of the following claims and their equivalents.

Claims

1. A touch display apparatus comprising: a display panel having a main display area and a first auxiliary display area outside the main display area; a plurality of first light-emitting elements arranged on the first auxiliary display area of the display panel; and a first conductive layer disposed on the display panel and comprising: a plurality of first mesh-shaped touch electrodes located on the main display area of the display panel; and a plurality of first dummy electrodes located on the main display area of the display panel, wherein each of the first dummy electrodes is disposed between two adjacent first mesh-shaped touch electrodes of the first mesh-shaped touch electrodes and structurally separated from the two adjacent first mesh-shaped touch electrodes, and at least one portion of the each of the first dummy electrodes is electrically connected to at least

one of the first light-emitting elements disposed in the first auxiliary display area.

2. The touch display apparatus according to claim 1, wherein the main display area and the first auxiliary display area are arranged in a first direction, each of the first mesh-shaped touch electrodes substantially extends in a first direction, the first mesh-shaped touch electrodes are arranged in a second direction intersecting with the first direction, the display panel further has a second auxiliary display area outside the main display area and the first auxiliary display area, the main display area and the second auxiliary display area are arranged in the second direction, and the touch display apparatus further comprises: a plurality of second light-emitting elements disposed on the second auxiliary display area; at least one insulation layer disposed on the first conductive layer; and a second conductive layer disposed on the at least one insulation layer and comprises: a plurality of second mesh-shaped touch electrodes disposed on the main display area of the display panel and interspersed with the first mesh-shaped touch electrodes; and a plurality of second dummy electrodes disposed on the main display area of the display panel, wherein each of the second dummy electrodes is disposed between two adjacent second mesh-shaped touch electrodes of the second mesh-shaped touch electrodes, the each of the second dummy electrodes and the two adjacent second mesh-shaped touch electrodes are structurally separated from each other, and at least a portion of the second dummy electrodes is electrically connected to at least one of the second light-emitting elements disposed in the second auxiliary display area.

3. The touch display apparatus according to claim 2, wherein the each of the first dummy electrodes comprises first conductive portions structurally separated from each other, the first conductive portions of the each of the first dummy electrodes are electrically connected to the first light-emitting elements disposed in the first auxiliary display area, the first conductive portions of the each of the first dummy electrodes define first dummy grids, and a shape of a first dummy grid of the first dummy grids is substantially the same as a shape of a first grid of the each of the first mesh-shaped touch electrodes.

4. The touch display apparatus according to claim 3, wherein a size of the first dummy grid is substantially the same as the size of the first grid of the each of the first mesh-shaped touch electrodes.

5. The touch display apparatus according to claim 3, wherein each of the first conductive portions comprises first line segments and second line segments, the first line segments and the second line segments are alternately arranged and connected in series, the first line segments are substantially parallel to each other and extend in a third direction, the first direction is substantially perpendicular to the second direction, the third direction intersects with the first direction and the second direction, the second line segments are substantially parallel to each other and extend in a fourth direction, and the fourth direction intersects with the first direction, the second direction and the third direction.

6. The touch display apparatus according to claim 3, wherein a first conductive portion of the first conductive portions of the each of the first dummy electrodes is electrically connected to first light-emitting elements of the first light-emitting elements.

7. The touch display apparatus according to claim 6, wherein first ends of the first light-emitting elements are electrically connected to first signal lines respectively, and second ends of the first light-emitting elements are electrically connected to the first conductive portion.
